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Dipole moment fluctuation formulas in computer simulations of polar systems

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The fundamental equation of electrostatics, i.e. the integral equation for the polarization of a macroscopic dielectric in an arbitrary external field, is written in a form that allows explicit inclusion of the toroidal boundary conditions as well as the cutoff of dipolar interactions often used in the computer simulation of polar systems. The toroidal boundary conditions are accounted for in a natural way if the integral equation is formulated (and solved) in Fourier space. Rigorous expressions for the polarization induced by a homogeneous field as well as for the equilibrium dipole moment fluctuations, both as a function of the dielectric constant, are then easily derived for general systems in two and three dimensions. The equations obtained for spherical cutoff geometry with reaction field are identical to those valid for an infinite periodic system (Ewald sum plus reaction field).

When applied to the case of a highly polar Stockmayer system ($\mu^{*2} = 3.0$, $\rho^* = 0.822$, $T^* = 1.15$) the formulas lead to a consistent value of the dielectric constant that is independent of the precise value of the reaction field—if the number of particles is sufficiently large. In the thermodynamic limit the dielectric constant obtained with the reaction field method seems to be much higher than one would expect from the Ewald sum results published for the Stockmayer system at the same thermodynamic state.

1. INTRODUCTION

Computer simulations of polar systems are usually performed in one of the following geometries (a pair of boundary conditions and interaction truncation scheme will be called a 'geometry'):

- (a) Toroidal boundary conditions with minimum image truncation of dipolar interactions (MI).
- (b) Toroidal boundary conditions with spherical interaction cutoff (SC).
- (c) Toroidal boundary conditions with spherical cutoff and approximation of neglected (as compared to an infinite system) interactions by the reaction field (RF)‡.

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‡ 'RF' will only be used as an abbreviation for reaction field *geometry* (c), as opposed to 'reaction field' in the general sense.

- (d) Periodic boundary conditions. Here interactions between particles in different cells of an infinite, though strictly periodic system are most often evaluated by the Ewald-Kornfeld or a similar lattice summation technique (EK)†.

In (a)–(d) we have deliberately used the terms ‘toroidal boundary conditions’ (TBC) and ‘periodic boundary conditions’ (PBC), in order to emphasize that the system being simulated is finite in cases (a)–(c) and infinite in case (d).

While SC is usually sufficient for systems with short-range interactions, the orientational correlations that are so important in polar systems, because of the long-range nature of dipolar interactions, are extremely sensitive to the type of geometry used in the simulation [1–12, 23, 35]. A well-known example is the equation relating the fluctuations of the total dipole moment $\mathbf{M}$ of a system to its dielectric constant ϵ . For an isolated spherical sample we have [13, 14]

$$\frac{\epsilon - 1}{\epsilon + 2} = \frac{4\pi}{3} \frac{\langle M^2 \rangle}{3Vk_B T}, \quad (1)$$

while, if the sample is embedded in an infinite medium of its own dielectric constant, we have [13–15]

$$\frac{(2\epsilon + 1)(\epsilon - 1)}{9\epsilon} = \frac{4\pi}{3} \frac{\langle M^2 \rangle}{3Vk_B T}. \quad (2)$$

Here V and T are volume and temperature of the sample; k_B is Boltzmann’s constant. (1) is a Clausius–Mossotti (CM) type equation; (2) is known as the Kirkwood–Fröhlich (KF) equation. Both are quoted in gaussian units which are also used throughout this article; this is done in order to facilitate comparison of results in two and three dimensions.

Referring to an early paper by Rahman and Stillinger [16] it has been argued that (1) should also hold for computer simulations in SC [5, 17] and MI-geometry [5, 18], while (2) is assumed to apply for RF-geometry [2, 5, 9, 10, 19, 20]. Although CM and KF have been used to interpret simulation results for almost a decade, a rigorous proof for the applicability of (1) and (2) in computer simulations involving a cubical sample shape, toroidal boundary conditions, and a finite interaction range has yet to be given. This state of affairs is illustrated by the confusion that prevailed for some time, as to whether the volume entering (1) and (2) is that of the cutoff sphere or the whole sample [4, 5, 9–11, 17, 18, 20–22]. In a recent paper Patey *et al.* [23] have derived a formula for general RF-geometry that supports (1) and (2). They have also pointed out that the $\langle M^2 \rangle$ and V to be used in their equation are those of the total sample. Although their formula is correct—it is identical with the one derived below—their proof includes neither the cubical sample shape nor the TBC explicitly.

Of course it is always possible to perform simulations in a geometry for which either CM or KF is known to hold, that is, one might try to simulate a spherical sample embedded in an infinite dielectric (as suggested by Friedman [24]) or a spherical sample surrounded by vacuum. The latter possibility

† ‘EK’ will be used for any one of these summation methods. That is, we would also call Ladd’s [6, 39] technique an EK-geometry.

has been followed by Bossis and coworkers [25–28]. While these alternatives certainly remove some ambiguities they nevertheless destroy the homogeneity of the system by necessitating the introduction of walls.

In this article we will therefore give a rigorous, formal and straightforward derivation of general fluctuation formulas for toroidal boundary conditions. This renewed interest in TBC seems to be the more justified in the light of theoretical papers by de Leeuw *et al.* [29, 30], Felderhof [31, 32] and Smith [33], where exact fluctuation formulas for EK-geometry are given. In particular it is evident from [29, 30] and [33] that some kind of reaction field *has* to be used even in EK-geometry, if the system is not to obey CM. As will be shown below, the fluctuation formulas for RF-geometry are the same as those for EK with reaction field. Thus the inclusion of the reaction field seems to be more important for an adequate simulation of polar systems than whether the system is finite (TBC) or infinite (PBC). Which one of these geometries is to be ultimately preferred (for a given number of particles RF would be less time-consuming and easier to program than EK) can only be clarified by an empirical investigation of the N -dependence of the dielectric constant.

The formulas necessary for an unambiguous interpretation of RF-simulations in two and three dimensions will be derived below. Their validity and applicability is then examined in a series of simulations, in various toroidal geometries (MI, SC, RF), of a highly polar Stockmayer system ($\mu^{*2} = 3.0$, $\rho^* = 0.822$, $T^* = 1.15$) for which results under PBC are available in the literature [10].

2. FLUCTUATION FORMULAS FOR TOROIDAL BOUNDARY CONDITIONS

When deriving fluctuation formulas by the Kirkwood–Fröhlich method [13–15] one usually proceeds in two steps: First, a relation is established, with the aid of statistical mechanics, between the mean total dipole moment $\langle \mathbf{M} \rangle_{E_0}$ of the sample V in the presence of a homogeneous external field $\mathbf{E}_0$, and the equilibrium fluctuations $\langle M^2 \rangle$ in the absence of any external perturbations. In the second step, classical electrostatics is used to calculate the macroscopic polarization $\mathbf{P}$ induced by $\mathbf{E}_0$ in the sample. After eliminating $\mathbf{E}_0$ from both equations one obtains a relation between $\langle M^2 \rangle$ and the dielectric constant.

2.1. The relation between $\langle \mathbf{M} \rangle_{E_0}$ and $\langle M^2 \rangle$

We will only briefly sketch the idea. A detailed derivation can be found in [13–15].

For a system of N particles with generalized coordinates $q^N = (q_1, \dots, q_N)$ and interaction energy $U(q^N)$, enclosed in a volume V at temperature T , the mean value of the total dipole moment in the direction of a homogeneous external field $\mathbf{E}_0$,

$$\langle \mathbf{M} \rangle_{E_0} = \frac{\int dq^N \exp \{ - [U(q^N) - \mathbf{M}(q^N) \cdot \mathbf{E}_0] / k_B T \} \mathbf{M}(q^N)}{\int dq^N \exp \{ - [U(q^N) - \mathbf{M}(q^N) \cdot \mathbf{E}_0] / k_B T \}}, \quad (3)$$

to first order in $\mathbf{E}_0$ is given by

$$\langle \mathbf{M} \rangle_{E_0} = \left\{ \begin{array}{l} \frac{\langle M^2 \rangle}{2k_B T} \mathbf{E}_0 \end{array} \right. \quad (4.2D)$$

$$\langle \mathbf{M} \rangle_{E_0} = \left\{ \begin{array}{l} \frac{\langle M^2 \rangle}{3k_B T} \mathbf{E}_0. \end{array} \right. \quad (4.3D)$$

In (4) use has been made of $\langle M_x^2 \rangle = \langle M_y^2 \rangle = \frac{1}{2} \langle M^2 \rangle$ and $\langle M_x^2 \rangle = \langle M_y^2 \rangle = \langle M_z^2 \rangle = \frac{1}{3} \langle M^2 \rangle$ in 2D and 3D, respectively, which is true regardless of whether the sample is spherical, cubical or toroidal. Since there is no distinction between interior and exterior in TBC, the external field appearing in (3) and (4) is $\mathbf{E}_0$ itself, rather than a cavity field, as in (7.2) of [13].

2.2. The relation between $\mathbf{P}$ and $\mathbf{E}_0$

In the presence of an external field $\mathbf{E}_0$ a non-vanishing polarization (dipole moment density) $\mathbf{P}$ will be induced in the sample which may be either a cube V (in 3D) or a square A (in 2D). According to electrostatics the total field at any point $\mathbf{r}$ in the sample is given by

$$\mathbf{E}(\mathbf{r}) = \mathbf{E}_0(\mathbf{r}) + \int d\mathbf{r}' \mathbf{T}(\mathbf{r} - \mathbf{r}') \cdot \mathbf{P}(\mathbf{r}'). \quad (5)$$

$\mathbf{T}$ is the dipole-dipole tensor, i.e.

$$T_{ab} = \begin{cases} \nabla_a \nabla_b (-\ln r) & (6.2D) \\ \nabla_a \nabla_b (1/r). & (6.3D) \end{cases}$$

Note that (5) is precisely the way in which the electric field would be calculated in a simulation, except that the integral would be replaced by a sum over particles.

With the aid of the constitutive relation $\mathbf{P}(\mathbf{r}) = \chi \mathbf{E}(\mathbf{r})$, with $\chi = (\epsilon - 1)/2\pi$ in 2D and $\chi = (\epsilon - 1)/4\pi$ in 3D, $\mathbf{E}$ may now be eliminated from (5), giving an integral equation for $\mathbf{P}$

$$\mathbf{P}(\mathbf{r}) = \chi \{ \mathbf{E}_0(\mathbf{r}) + \int d\mathbf{r}' \mathbf{T}(\mathbf{r} - \mathbf{r}') \cdot \mathbf{P}(\mathbf{r}') \}. \quad (7)$$

Since $\mathbf{T}$ is singular at the origin it is convenient to split the integral on the right hand side into two parts by integrating either over $|\mathbf{r} - \mathbf{r}'| < \rho$ or $|\mathbf{r} - \mathbf{r}'| > \rho$ and letting $\rho \rightarrow 0$. For the part containing the singularity one obtains in 3D [13–15]

$$\lim_{\rho \rightarrow 0} \int_{|\mathbf{r} - \mathbf{r}'| < \rho} d\mathbf{r}' \mathbf{T}(\mathbf{r} - \mathbf{r}') \cdot \mathbf{P}(\mathbf{r}') = -\frac{4\pi}{3} \mathbf{P}(\mathbf{r}). \quad (8.3D)$$

Equation (8) may also be derived explicitly by substituting the definition (6.3D) on the left hand side and rewriting the divergence as a surface integral. In the limit $\rho \rightarrow 0$ one then arrives at (8). The fastest way to find the 2D analogue is to formally replace the 3D volume of the unit sphere ($4\pi/3$) by the 2D area of the unit circle (π)

$$\lim_{\rho \rightarrow 0} \int_{|\mathbf{r} - \mathbf{r}'| < \rho} d\mathbf{r}' \mathbf{T}(\mathbf{r} - \mathbf{r}') \cdot \mathbf{P}(\mathbf{r}') = -\pi \mathbf{P}(\mathbf{r}). \quad (8.2D)$$

Thus the integral equation for $\mathbf{P}$ reads

$$\mathbf{P}(\mathbf{r}) = \lambda \left\{ \mathbf{E}_0(\mathbf{r}) + \lim_{\rho \rightarrow 0} \int_{|\mathbf{r} - \mathbf{r}'| > \rho} d\mathbf{r}' \mathbf{T}(\mathbf{r} - \mathbf{r}') \cdot \mathbf{P}(\mathbf{r}') \right\}. \quad (9)$$

For the sake of brevity we shall omit the limit as well as the domain of integration in the following. λ is an abbreviation for

$$\lambda = \begin{cases} \frac{1}{\pi} \frac{\epsilon - 1}{\epsilon + 1} & (10.2D) \\ \frac{3}{4\pi} \frac{\epsilon - 1}{\epsilon + 2} & (10.3D) \end{cases}$$

(9) and (10) are the basic equations of electrostatics. We have to solve them for the case of a homogeneous external field. Although (9) could easily be solved directly (for TBC $\mathbf{P}$ will be homogeneous too, if $\mathbf{E}_0$ is homogeneous), the method of solution best suited to an integral equation containing a convolution with toroidal boundary conditions is Fourier transformation. (This is illustrated for a one-dimensional example in the Appendix.)

We therefore write (9) in Fourier space,

$$\tilde{\mathbf{P}}(\mathbf{k}) = \lambda \{ \tilde{\mathbf{E}}_0(\mathbf{k}) + \tilde{\mathbf{T}}(\mathbf{k}) \cdot \tilde{\mathbf{P}}(\mathbf{k}) \}, \quad (11)$$

whose solution is

$$\tilde{\mathbf{P}}(\mathbf{k}) = \lambda [1 - \lambda \tilde{\mathbf{T}}(\mathbf{k})]^{-1} \cdot \tilde{\mathbf{E}}_0(\mathbf{k}). \quad (12)$$

Equation (12) is our central result. In a subsequent publication it will be used to calculate the distortions of the dipolar field of a particle that are created by various boundary conditions.

In the present case $\mathbf{E}_0$ is homogeneous, that is, if $E_{0,x} = E_{0,y} = 0$ and $E_{0,z} = E_0$ in 3D, we have

$$\left. \begin{aligned} \tilde{E}_{0,x}(\mathbf{k}) = \tilde{E}_{0,y}(\mathbf{k}) &= 0 \quad \text{for all } \mathbf{k}, \\ \tilde{E}_{0,z}(\mathbf{k}) &= \begin{cases} VE_0 & \text{for } \mathbf{k} = 0 \\ 0 & \text{otherwise.} \end{cases} \end{aligned} \right\} \quad (13)$$

Two typical components of $\mathbf{T}$ are $T_{xy} = 3xy/r^5$ and $T_{zz} = (3z^2 - r^2)/r^5$. Because of (13) it is sufficient to know $\tilde{\mathbf{T}}(\mathbf{k})$ for $\mathbf{k} = 0$, that is,

$$\tilde{\mathbf{T}}(0) = \int d\mathbf{r} \mathbf{T}(\mathbf{r}). \quad (14)$$

For MI-geometry the integral is over the whole cube, for SC-geometry it is only over a sphere of radius r_0 , corresponding to the finite range of the dipole-dipole interaction. Since the singular part of $\mathbf{T}$, $r < \rho$, has been accounted for explicitly in (9), we may argue that T_{xy} is an odd function and therefore $\tilde{T}_{xy}(0) = 0$, if T_{xy} is integrated over a cube or a sphere. Similarly, we have $\int d\mathbf{r} 3z^2/r^5 = \int d\mathbf{r}/r^3$ in both cases (cube and sphere), and therefore $\tilde{T}_{zz}(0) = 0$. Thus

$$\tilde{\mathbf{T}}(0) = 0 \quad (15)$$

for MI as well as SC-geometry.

(12) now simplifies to $\tilde{\mathbf{P}}(0) = \lambda \tilde{\mathbf{E}}_0(0)$, giving $M_z = VP_z = \tilde{P}_z(0) = \lambda VE_0$. Eliminating $\mathbf{E}_0$ from (4) we immediately conclude that CM, a Clausius–Mossotti-like equation, is valid for MI as well as SC-geometry

$$\frac{\epsilon - 1}{\epsilon + 1} = \pi \frac{\langle M^2 \rangle}{2Ak_B T}, \quad (16.2D)$$

$$\frac{\epsilon - 1}{\epsilon + 2} = \frac{4\pi}{3} \frac{\langle M^2 \rangle}{3Vk_B T}. \quad (16.3D)$$

From the above it is clear that A and V are the domain of integration used in the Fourier transforms, that is, area and volume, respectively, of the total system (rather than the cutoff sphere). The derivation of the 2D equation is completely analogous to the one given for 3D.

3. THE REACTION FIELD

Here those interactions of every particle i with its more distant neighbours j ($r_{ij} > r_c$) that are neglected in SC-geometry are replaced by an infinite continuum with dielectric constant ϵ_{RF} [2, 3]. In this way, it is hoped, one should be able to simulate an infinite system, provided $\epsilon_{\text{RF}} \approx \epsilon$. The total field acting on particle i now has to be corrected by the reaction field, in 3D

$$\mathbf{R}(i) = \frac{2(\epsilon_{\text{RF}} - 1)}{2\epsilon_{\text{RF}} + 1} \frac{1}{r_c^3} \mathbf{M}(i), \quad (17)$$

which is added to the field due to the immediate neighbours ($r_{ij} < r_c$) of i . In (17) $\mathbf{M}(i)$ is the total dipole moment of the cutoff sphere around i . Equivalently [8], one may use a modified dipole–dipole interaction

$$u_{\text{DD}}(ij) = \begin{cases} -\boldsymbol{\mu}(i) \cdot \mathbf{T}(ij) \cdot \boldsymbol{\mu}(j) - \frac{2(\epsilon_{\text{RF}} - 1)}{2\epsilon_{\text{RF}} + 1} \frac{\boldsymbol{\mu}(i) \cdot \boldsymbol{\mu}(j)}{r_c^3} & \text{if } r_{ij} < r_c \\ 0 & \text{otherwise.} \end{cases} \quad (18)$$

The way to write (18) that is best suited to our purposes is as $u_{\text{DD}}(ij) = -\boldsymbol{\mu}(i) \cdot \mathbf{T}(ij, \epsilon_{\text{RF}}) \cdot \boldsymbol{\mu}(j)$. The modified $\mathbf{T}$ -tensor is thus defined by

$$T_{ab}(ij, \epsilon_{\text{RF}}) = \begin{cases} T_{ab}(ij) + \frac{2(\epsilon_{\text{RF}} - 1)}{2\epsilon_{\text{RF}} + 1} \frac{1}{r_c^3} \delta_{ab} & \text{if } r_{ij} < r_c \\ 0 & \text{otherwise.} \end{cases} \quad (19.3D)$$

The corresponding expression in 2D is [11, 25, 33]

$$T_{ab}(ij, \epsilon_{\text{RF}}) = \begin{cases} T_{ab}(ij) + \frac{\epsilon_{\text{RF}} - 1}{\epsilon_{\text{RF}} + 1} \frac{1}{r_c^2} \delta_{ab} & \text{if } r_{ij} < r_c \\ 0 & \text{otherwise.} \end{cases} \quad (19.2D)$$

In the following we will use the abbreviation

$$\mu = \begin{cases} \frac{1}{\pi} \frac{\epsilon_{\text{RF}} - 1}{\epsilon_{\text{RF}} + 1} \end{cases} \quad (20.2\text{D})$$

$$\mu = \begin{cases} \frac{3}{4\pi} \frac{2(\epsilon_{\text{RF}} - 1)}{2\epsilon_{\text{RF}} + 1} \end{cases} \quad (20.3\text{D})$$

The derivation of fluctuation formulas is now completely analogous to the case of SC, the only difference being that the integral

$$\tilde{\mathbf{T}}(\mathbf{k}, \epsilon_{\text{RF}}) = \int d\mathbf{r} \exp(-i\mathbf{k} \cdot \mathbf{r}) \mathbf{T}(\mathbf{r}, \epsilon_{\text{RF}})$$

does not vanish for $\mathbf{k} = 0$. One rather has

$$\tilde{T}_{ab}(0, \epsilon_{\text{RF}}) = \begin{cases} \pi^2 \mu \delta_{ab} \end{cases} \quad (21.2\text{D})$$

$$\tilde{T}_{ab}(0, \epsilon_{\text{RF}}) = \begin{cases} \left(\frac{4\pi}{3}\right)^2 \mu \delta_{ab} \end{cases} \quad (21.3\text{D})$$

Combining (21) and (12) yields, in 3D,

$$\begin{aligned} \tilde{\mathbf{P}}(0) &= \lambda [1 - \lambda \tilde{\mathbf{T}}(0, \epsilon_{\text{RF}})]^{-1} \cdot \mathbf{E}_0(0) \\ &= \lambda \left[1 - \left(\frac{4\pi}{3}\right)^2 \lambda \mu \right]^{-1} \mathbf{E}_0(0) \end{aligned} \quad (22)$$

or

$$M_z = \lambda V E_0 \left/ \left[1 - \left(\frac{4\pi}{3}\right)^2 \lambda \mu \right] \right. \quad (23)$$

Eliminating $\mathbf{E}_0$ from (4) and substituting for λ and μ we obtain

$$\frac{\epsilon - 1}{\epsilon + 2} \left[1 - \frac{\epsilon - 1}{\epsilon + 2} \frac{2(\epsilon_{\text{RF}} - 1)}{2\epsilon_{\text{RF}} + 1} \right]^{-1} = \frac{4\pi}{3} \frac{\langle M^2 \rangle}{3V k_B T} \quad (24.3\text{D})$$

The result for 2D is

$$\frac{\epsilon - 1}{\epsilon + 1} \left[1 - \frac{\epsilon - 1}{\epsilon + 1} \frac{\epsilon_{\text{RF}} - 1}{\epsilon_{\text{RF}} + 1} \right]^{-1} = \pi \frac{\langle M^2 \rangle}{2A k_B T} \quad (24.2\text{D})$$

An alternative method to find the dielectric constant of a model system is to measure the response to a homogeneous external field $\mathbf{E}_0$, i.e. the polarization induced by that field [9, 10, 19, 22, 34]. Equation (23) immediately leads to

$$\pi \langle \mathbf{P} \rangle_{\mathbf{E}_0} = \frac{\epsilon - 1}{\epsilon + 1} \left[1 - \frac{\epsilon - 1}{\epsilon + 1} \frac{\epsilon_{\text{RF}} - 1}{\epsilon_{\text{RF}} + 1} \right]^{-1} \mathbf{E}_0, \quad (25.2\text{D})$$

$$\frac{4\pi}{3} \langle \mathbf{P} \rangle_{\mathbf{E}_0} = \frac{\epsilon - 1}{\epsilon + 2} \left[1 - \frac{\epsilon - 1}{\epsilon + 2} \frac{2(\epsilon_{\text{RF}} - 1)}{2\epsilon_{\text{RF}} + 1} \right]^{-1} \mathbf{E}_0. \quad (25.3\text{D})$$

$\langle \mathbf{P} \rangle_{\mathbf{E}_0}$ is the polarization induced by, and parallel to, the external field.

4. FLUCTUATION AND POLARIZATION FORMULAS

Equations (24) and (25) of the preceding section are quite general in the sense that they permit the evaluation of the dielectric constant for arbitrary toroidal geometry. The formulas applicable to MI and SC-geometry are obtained by formally letting $\epsilon_{\text{RF}} \rightarrow 1$ in (24) and (25). Because of their importance some special cases are listed below.

4.1. Fluctuation formulas

Two dimensions

$$\pi \frac{\langle M^2 \rangle}{2Ak_{\text{B}}T} = \begin{cases} \frac{\epsilon-1}{\epsilon+1} \left[1 - \frac{\epsilon-1}{\epsilon+1} \frac{\epsilon_{\text{RF}}-1}{\epsilon_{\text{RF}}+1} \right]^{-1} & \text{RF} & (26.1.2\text{D}) \\ \frac{\epsilon-1}{\epsilon+1} & \text{MI/SC} & (26.2.2\text{D}) \\ \frac{(\epsilon+1)(\epsilon-1)}{4\epsilon} & \epsilon_{\text{RF}} = \epsilon & (26.3.2\text{D}) \\ \frac{\epsilon-1}{2} & \epsilon_{\text{RF}} = \infty. & (26.4.2\text{D}) \end{cases}$$

Three dimensions

$$\frac{4\pi}{3} \frac{\langle M^2 \rangle}{3Vk_{\text{B}}T} = \begin{cases} \frac{\epsilon-1}{\epsilon+2} \left[1 - \frac{\epsilon-1}{\epsilon+2} \frac{2(\epsilon_{\text{RF}}-1)}{2\epsilon_{\text{RF}}+1} \right]^{-1} & \text{RF} & (26.1.3\text{D}) \\ \frac{\epsilon-1}{\epsilon+2} & \text{MI/SC} & (26.2.3\text{D}) \\ \frac{(2\epsilon+1)(\epsilon-1)}{9\epsilon} & \epsilon_{\text{RF}} = \epsilon & (26.3.3\text{D}) \\ \frac{\epsilon-1}{3} & \epsilon_{\text{RF}} = \infty. & (26.4.3\text{D}) \end{cases}$$

As stated in the introduction, it is possible to prove rigorously that a Clausius-Mossotti-like equation actually holds for MI and SC-geometry while the Kirkwood-Fröhlich equation is correct for RF-geometry with $\epsilon_{\text{RF}} = \epsilon$. In contrast to [23] these equations are seen to be valid even for a cubical (square) volume (area) and toroidal boundary conditions. The volume (area) to be inserted on the left hand side of (26) in SC and RF-geometry is that of the total sample. This has been pointed out in [23] too. The formulas are in fact independent of r_{e} .

We also notice that (26) is identical to the formulas known to hold for EK-geometry with reaction field [29, 30, 33]. In particular (26.4) is just the formula for 'conducting boundary conditions'

$$\epsilon = \begin{cases} 1 + \pi \frac{\langle M^2 \rangle}{Ak_{\text{B}}T} & (27.2\text{D}) \\ 1 + \frac{4\pi}{3} \frac{\langle M^2 \rangle}{Vk_{\text{B}}T} & (27.3\text{D}) \end{cases}$$

We further notice that, in contrast to what one might expect, it makes quite a difference to replace ϵ_{RF} by ∞ in (17), even for systems with large ϵ ; i.e. to replace $(\epsilon_{\text{RF}} - 1)/(\epsilon_{\text{RF}} + 1)$ or $2(\epsilon_{\text{RF}} - 1)/(2\epsilon_{\text{RF}} + 1)$ by unity: if (26.3) were used instead of (26.4) to extract the dielectric constant from a simulation with $\epsilon_{\text{RF}} = \infty$, a value of ϵ would result that is too large by a factor of 2 (in 2D) or 3/2 (in 3D). Thus it proves essential to work with the correct formula.

4.2. Polarization formulas

Two dimensions

$$\pi \langle P \rangle_{E_0} / E_0 = \begin{cases} \frac{\epsilon - 1}{\epsilon + 1} \left[1 - \frac{\epsilon - 1}{\epsilon + 1} \frac{\epsilon_{\text{RF}} - 1}{\epsilon_{\text{RF}} + 1} \right]^{-1} & \text{RF} & (28.1.2D) \\ \frac{\epsilon - 1}{\epsilon + 1} & \text{MI/SC} & (28.2.2D) \\ \frac{(\epsilon + 1)(\epsilon - 1)}{4\epsilon} & \epsilon_{\text{RF}} = \epsilon & (28.3.2D) \\ \frac{\epsilon - 1}{2} & \epsilon_{\text{RF}} = \infty. & (28.4.2D) \end{cases}$$

Three dimensions

$$\frac{4\pi}{3} \langle P \rangle_{E_0} / E_0 = \begin{cases} \frac{\epsilon - 1}{\epsilon + 2} \left[1 - \frac{\epsilon - 1}{\epsilon + 2} \frac{2(\epsilon_{\text{RF}} - 1)}{2\epsilon_{\text{RF}} + 1} \right]^{-1} & \text{RF} & (28.1.3D) \\ \frac{\epsilon - 1}{\epsilon + 2} & \text{MI/SC} & (28.2.3D) \\ \frac{(2\epsilon + 1)(\epsilon - 1)}{9\epsilon} & \epsilon_{\text{RF}} = \epsilon & (28.3.3D) \\ \frac{\epsilon - 1}{3} & \epsilon_{\text{RF}} = \infty. & (28.4.3D) \end{cases}$$

Again (28.4) is the formula for conducting boundary conditions

$$\epsilon = \begin{cases} 1 + \pi \langle P \rangle_{E_0} / E_0 & (29.2D) \\ 1 + 4\pi \langle P \rangle_{E_0} / E_0. & (29.3D) \end{cases}$$

E_0 in (28) and (29) is the field actually added to the interaction of each particle, since this is how E_0 has been defined in (5). The total field, including the contribution from the polarization of the system, is of course linked to $\mathbf{P}$ by the constitutive relation $\mathbf{P} = (\epsilon - 1)/4\pi \mathbf{E}$, but generally not known in advance. In fact, knowledge of the correct relationship between E_0 and $\mathbf{E}$ is essential for a successful interpretation of applied field simulations.

As an illustration, we consider the analysis by Adams and Adams [10] of their simulations in RF-geometry: Assuming that RF corresponds to a spherical sample embedded in an infinite medium with dielectric constant ϵ_{RF} , Adams

and Adams used the appropriate cavity field $3\epsilon_{\text{RF}}/(2\epsilon_{\text{RF}} + 1)\mathbf{E}$ as the $\mathbf{E}_0$ (in our notation) in their simulations. The dielectric constant was then determined from the constitutive relation

$$\epsilon = 1 + 4\pi\langle P \rangle / E. \quad (28.3.3D')$$

Inserting Adams and Adams' $\mathbf{E}_0$ into (28.1.3D), however, shows that

$$\epsilon = 1 + \frac{2\epsilon_{\text{RF}} + \epsilon}{3\epsilon_{\text{RF}}} 4\pi\langle P \rangle / E \quad (28.1.3D')$$

should have been used instead. Although (28.3.3D') is correct for $\epsilon \approx \epsilon_{\text{RF}}$, it will give a value for the dielectric constant that is too low if $\epsilon_{\text{RF}} < \epsilon$, and too high if $\epsilon_{\text{RF}} > \epsilon$ [41].

4.3. Remarks

From (26)–(29) it is evident that, regardless of whether the equilibrium fluctuations of the total dipole moment or the polarization in an applied field is used to calculate the dielectric constant, for highly polar systems only simulations including the reaction field can be successful: $\langle M^2 \rangle$ and $\langle P \rangle_{E_0}/E_0$ are both bounded in the limit $\epsilon \rightarrow \infty$ for MI as well as SC-geometry (and also for EK without reaction field), but proportional to ϵ for RF-geometry (or EK with reaction field). Because of the limited statistical accuracy of simulations it will only be possible in the latter case to arrive at a reasonable value for the dielectric constant.

In order to remove the discontinuity introduced by the potential cutoff, in some simulations [10, 12, 20, 35] the dipole–dipole tensor (the modified $\mathbf{T}$ -tensor if the reaction field is used) is multiplied by a smoothing function $f(\mathbf{r})$, where $f(0)=1$, $f(\mathbf{r})=0$ for $r > r_c$, and $f(\mathbf{r})\mathbf{T}(\mathbf{r})$ is a continuously differentiable function. If $\mathbf{T}$ is replaced by $f\mathbf{T}$ throughout the derivation given in § 2, the formulas valid for MI and SC-geometry are found to apply to this case too: Since $f(\mathbf{r})$ usually has spherical or cubical symmetry one has $\int d\mathbf{r} f(\mathbf{r})\mathbf{T}(\mathbf{r})=0$, the only assumption needed in § 2. In RF-geometry, on the other hand, where $\mathbf{T}$ as well as the reaction field is modulated by $f(\mathbf{r})$, one just has to replace πr_c^2 or $4\pi r_c^3/3$ by

$$\int_{r < r_c} d\mathbf{r} f(\mathbf{r})$$

in (17)–(19). In this way (21) and eventually (24) and (25) are recovered again. This is the recipe suggested by Adams *et al.* [20] which now has been justified. Failing to do so means that one is working with the reaction field corresponding to a continuum with dielectric constant ϵ'_{RF} , where ϵ'_{RF} is given by

$$\frac{2(\epsilon'_{\text{RF}} - 1)}{2\epsilon'_{\text{RF}} + 1} \frac{4\pi/3}{\int_{r < r_c} d\mathbf{r} f(\mathbf{r})} = \frac{2(\epsilon_{\text{RF}} - 1)}{2\epsilon_{\text{RF}} + 1} \frac{1}{r_c^3}. \quad (30)$$

This is the value (generally $\epsilon'_{\text{RF}} < \epsilon_{\text{RF}}$) to be used in the analysis of the simulation results [12].

One might think that it is possible to use the reaction field in MI-geometry too: (21) remains unchanged if r_c^2 and r_c^3 in (17–19) are replaced by A/π and $3V/4\pi$, respectively. In this way (24) and (25) are obtained once more. While this is certainly true it can be shown [37] that the reaction field, when used in

connection with MI-geometry, is not able to suppress, to the same degree as the reaction field used with a spherical cutoff, the influence of toroidal boundary conditions on the *local* (nearest neighbour) orientational structure of the system. Thus RF-geometry is to be preferred to MI in any case (with or without reaction field).

5. THE DIELECTRIC CONSTANT OF A HIGHLY POLAR STOCKMAYER SYSTEM

In order to test the applicability and consistency of (26) in actual computer simulations, a typical model system has been studied under various geometries at a single thermodynamic state. The system chosen is the 3D Stockmayer system with reduced dipole moment $\mu^{*2}=3.0$, at reduced density $\rho^*=0.822$ and temperature $T^*=1.15$ ($\mu^{*2}=\mu^2/\epsilon\sigma^3$, $\rho^*=\rho\sigma^3$, and $T^*=k_B T/\epsilon$, where ϵ and σ are the parameters of the Lennard-Jones part of the potential)[†]. The same system has also been investigated by Adams and Adams [10] in some detail, although mostly in EK-geometry. The simulations reported for RF-geometry in [10] have been analysed using $\langle M^2 \rangle$ and V of the cutoff sphere and are not useful for our purposes.

The results reported below have been obtained from molecular dynamics (MD) simulations of a system of 256 particles. For the integration of the equations of motion the Singer-Verlet algorithm [38] with a (reduced) time step of $\Delta t^*=0.0025$ was used. The moment of inertia (which has no effect on the purely static properties that are of sole interest here) was taken to be $I^*=0.025$. With that system a simulation was performed each in MI, SC and RF (with $\epsilon_{RF}=50$ and $\epsilon_{RF}=\infty$) geometry. The potential cutoff in the latter simulations was always half the box length, i.e. $r_c=L/2=3.389\sigma$ at that density. No smoothing function of the kind described at the end of § 4 was used. In order to compensate for the increase in energy that is to be expected in that case [20], the total energy of the system was artificially kept constant by rescaling the translational and rotational velocities after every 100 time steps. (The energy increase typically observed in that interval was of the order of 0.25–0.5 per cent, depending on boundary conditions.)

For the simulations in RF-geometry the desired statistical accuracy of 5 per cent for $\langle M^2 \rangle$ (and thus for ϵ) was reached after 50 000 time steps. This is in agreement with [34]. The runs in MI and SC-geometry were terminated after 20 000 Δt since, for a system of only 256 particles, it would be impossible to obtain a reasonable value for the dielectric constant even after a much longer time (see discussion of figure 2). The number of time steps typically needed to equilibrate the system was 10 000. This may be interpreted as the time the system needs to readjust itself to a change of geometry.

In the table some properties of the Stockmayer system are listed for various geometries and compared with Adams and Adams' [10] results. The values of special interest are those of $\langle M^2 \rangle$, G_K , g_K and ϵ . Here g_K is the Kirkwood g -factor, defined in the usual way [13–15] (i.e. for an infinite system); G_K is

[†] The system studied here is a singular case in the sense that, for a given polarity, its orientational correlations are of much longer range than those of a system with a more realistic interaction potential [12, 21, 35, 36].

Thermodynamic and dielectric properties of a Stockmayer system with $\mu^* = 3.0$ at $\rho^* = 0.822$.

Geometry	N	T^*	$\langle U^* \rangle / N$	$\langle U_{\text{DD}}^* \rangle / N$	$\langle p^* \rangle$	$G_K = \langle M^2 \rangle / N \mu^2$	g_K	ϵ	Time steps/ method †	Source
MI	108	1.15	-10.53	-5.25	-0.512	0.28	—	—	10 000 MC	[10]
MI	256	1.155	-10.430	-5.210	-0.551	0.29	—	—	20 000 MD	‡
SC	108	1.15	-9.97	-4.78	-0.085	0.32	—	—	10 000 MC	[10]
SC	256	1.148	-9.970	-4.777	-0.088	0.31	—	—	20 000 MD	‡
SC	500	1.15	-10.04	-4.85	-0.085	0.28	—	—	2000 MC	[10]
RF	$\left\{ \begin{array}{l} 108 \\ 256 \end{array} \right\}$	$\left\{ \begin{array}{l} 1.148 \\ 1.150 \end{array} \right\}$	$\left\{ \begin{array}{l} -9.834 \\ -9.896 \end{array} \right\}$	$\left\{ \begin{array}{l} -4.664 \\ -4.722 \end{array} \right\}$	$\left\{ \begin{array}{l} 0.100 \\ 0.057 \end{array} \right\}$	$\left\{ \begin{array}{l} 3.56 \\ 4.41 \end{array} \right\}$	$\left\{ \begin{array}{l} 3.51 \\ 4.87 \end{array} \right\}$	$\left\{ \begin{array}{l} 47.9 \\ 66.1 \end{array} \right\}$	$\left\{ \begin{array}{l} 100\,000\text{ MD} \\ 50\,000\text{ MD} \end{array} \right\}$	$\left\{ \begin{array}{l} ‡ \\ ‡ \end{array} \right\}$
RF	$\left\{ \begin{array}{l} 108 \\ 256 \end{array} \right\}$	$\left\{ \begin{array}{l} 1.149 \\ 1.154 \end{array} \right\}$	$\left\{ \begin{array}{l} -9.856 \\ -9.906 \end{array} \right\}$	$\left\{ \begin{array}{l} -4.692 \\ -4.730 \end{array} \right\}$	$\left\{ \begin{array}{l} 0.119 \\ 0.062 \end{array} \right\}$	$\left\{ \begin{array}{l} 5.78 \\ 7.09 \end{array} \right\}$	$\left\{ \begin{array}{l} 3.89 \\ 4.77 \end{array} \right\}$	$\left\{ \begin{array}{l} 53.0 \\ 64.5 \end{array} \right\}$	$\left\{ \begin{array}{l} 100\,000\text{ MD} \\ 50\,000\text{ MD} \end{array} \right\}$	$\left\{ \begin{array}{l} ‡ \\ ‡ \end{array} \right\}$
EK	108	1.15	-9.935	-4.737	-0.028	(4.29)§	(2.89)	39.5	50 000 MC	[10]
EK	256	1.15	-9.922	—	—	(3.48)	(2.36)	32.3	12 000 MC	[10]
EK	500	1.15	-9.902	-4.721	-0.020	(4.45)	(3.01)	41.0	20 000 MC	[10]
EK	500	1.155	-9.888	-4.705	0.056	(5.34)	(3.60)	48.8	30 000 MD	[10]

† For Monte Carlo simulations : order of magnitude of (number of moves)/ N .‡ This work. Estimated accuracy of G_K is 5 per cent. LJ energy and pressure include a cutoff correction. The LJ cutoff was always at $r_c = L/2$, even for MI. The electrostatic self-energy for RF-geometry, $\langle U_c^* \rangle / N = -(\mu^*/r_c^*)^3(\epsilon_{\text{RF}} - 1)/(2\epsilon_{\text{RF}} + 1)$, is not included in $\langle U^* \rangle / N$, $\langle U_{\text{DD}}^* \rangle / N$ and $\langle p^* \rangle$.

§ The numbers in parentheses have been calculated from the dielectric constants given in [10].

the g -factor for a finite system [8], a cube with toroidal boundary conditions in our case. Thus, if $\mu(i)$ is the dipole moment of particle i ,

$$G_K = \left\langle \sum_{j=1}^N \mu(i) \cdot \mu(j) \right\rangle / \mu^2 = \langle M^2 \rangle / N\mu^2. \quad (31)$$

Since G_K coincides with g_K for $\epsilon_{RF} = \epsilon$, and because of (26), we have, for arbitrary reaction field, in 3D

$$g_K = G_K \frac{(2\epsilon + 1)(\epsilon - 1)}{9\epsilon} \left[1 - \frac{\epsilon - 1}{\epsilon + 2} \frac{2(\epsilon_{RF} - 1)}{2\epsilon_{RF} + 1} \right]. \quad (32)$$

(Cf. also (14) of [16].) This is the way the g_K -values in the table have been obtained; the values for the dielectric constant have been calculated with the aid of (26).

According to (26.2.3D) G_K is bounded for $\epsilon \rightarrow \infty$ in MI and SC-geometry:

$$G_{K,MI/SC} < \frac{9k_B T}{4\pi\sigma\mu^2} = 0.334. \quad (33)$$

Although this inequality is obeyed (our values are also identical with those of [10]), the ϵ -values for MI and SC-geometry are neither in agreement with one another nor with those calculated for RF-geometry.

If ϵ is determined in RF-geometry, however, the result is even independent of ϵ_{RF} (unless $\epsilon_{RF} \ll \epsilon$)—provided the correct formula, namely (26.1), is used. We thus obtain $\epsilon = 66$ for $\epsilon_{RF} = 50$ and $\epsilon = 65$ for $\epsilon_{RF} = \infty$, a consistent result. Yet this value is significantly higher than that reported by Adams and Adams ($\epsilon \approx 50$) for EK-geometry†. In principle one would expect RF and EK to give consistent results in the limit $N \rightarrow \infty$. Whether the discrepancy is due to a different N -dependence for both geometries remains to be seen. Given our computational facilities, a system of 256 was close to the limit of what can be done in a reasonable space of time. Some results obtained for a smaller system ($N = 108$, $\epsilon_{RF} = 50$ and $\epsilon_{RF} = \infty$, 100 000 Δt each) are also included in the table. Here the ϵ -values are in closer agreement with those of [10]; they are also fairly consistent among themselves. However, preliminary results for a system of 500 particles seem to indicate that, in the thermodynamical limit, the dielectric constant is identical with that for $N = 256$. The EK-values, on the other hand, increase much less with N : they are not even monotonous.

In this context it has to be pointed out once more (see also the discussion following (27)), how sensitive the fluctuations of the total dipole moment are to any change of the precise form of the dipole-dipole interaction. It might not be impossible that the approximations necessary for a full evaluation of the Ewald sum systematically distort the results‡. Ladd's [6, 39] multipole expansion of the Ewald sum, if augmented by the reaction field, would be conceptually more attractive: it can be shown [37] that the $\mathbf{k} = 0$ component of the Fourier transform vanishes for the individual terms in the series expansion of the generalized $\mathbf{T}$ -tensor (cf. (15)). Thus (26) is *exactly* obeyed, no matter where the series is truncated.

† Adams and Adams' EK-result is also the reason why $\epsilon_{RF} = 50$, rather than $\epsilon_{RF} = \epsilon \approx 65$, was used in one of our simulations. They were run in parallel.

‡ Note added in proof.—We have recently been able to show that this is indeed the case [42].

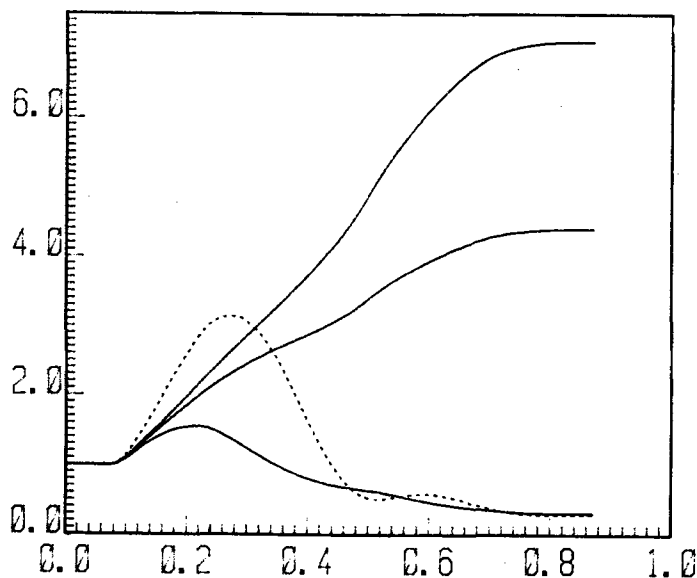


Figure 1. $\langle M^2(R) \rangle / \langle N(R) \rangle \mu^2$, the dipole moment fluctuations of a spherical volume with radius R , for a system of 256 Stockmayer particles with $\mu^{*2} = 3.0$ at $\rho^* = 0.822$ and $T^* = 1.15$. $N(R)$ is the number of particles in $V = 4\pi R^3/3$. For $R > L/2$, V is the intersection of a sphere of radius R with the MD cell. R is in units of the box length L . Dashed curve: MI-geometry; solid curves (top to bottom): RF with $\epsilon_{RF} = \infty$, RF with $\epsilon_{RF} = 50$, and SC-geometry.

For a dielectric sphere embedded in an infinite system, $\langle M^2 \rangle / V$ should be independent of the radius of that sphere, according to (12). In figure 1 $\langle M^2(R) \rangle / \langle N(R) \rangle \mu^2$ is plotted as a function of R . Here

$$\langle M^2(R) \rangle = \left\langle \left(\sum_{|\mathbf{r}_i - \mathbf{r}_0(t)| < R} \boldsymbol{\mu}(i) \right)^2 \right\rangle \quad (34)$$

and

$$\langle N(R) \rangle = \left\langle \sum_{|\mathbf{r}_i - \mathbf{r}_0(t)| < R} 1 \right\rangle. \quad (35)$$

$\mathbf{r}_0(t)$ is a point somewhere in the sample, chosen at random for every time step, and not necessarily coincident with the position of a particle. Thus $M(R)$ and $N(R)$ are the total dipole moment of, and the number of particles within, a sphere (or that part of a sphere that is within the cube-shaped sample) of radius R around $\mathbf{r}_0$. From figure 1 it is evident that it is impossible to find a reasonable value for the dielectric constant in MI or SC-geometry, using KF and the dipole moment fluctuations of a sphere that is small compared to the system as a whole: the resulting ϵ would be too small—at least for a system of our size. If CM is used instead, a negative value for ϵ is the most likely outcome [22]: $\langle M^2(R) \rangle / \langle N(R) \rangle \mu^2$ is everywhere greater than the maximum value permitted by (33). In RF-geometry, on the other hand, it is an increasing function of R , and ϵ can only be obtained from $\langle M^2 \rangle$ for the total volume (and not from the cutoff sphere). For $R/L \rightarrow \sqrt{3}/2$ the curves in figure 1 approach the $\langle M^2 \rangle$ -values given in the table.

The R -dependent Kirkwood g -factor [8], the total dipole moment of a sphere of radius R around a reference particle,

$$G_K(R) = \left\langle \sum_{j, r_{ij} < R} \mu(i) \cdot \mu(j) \right\rangle / \mu^2, \quad (36)$$

is plotted in figure 2. As before, the limits for $R/L \rightarrow \sqrt{3}/2$ are identical with the G_K -values in the table. The curve for SC-geometry shows the familiar pattern, a positive and a negative region, that is known to be caused by the boundary conditions [8]. For MI-geometry the situation is similar, with positive values being increased and negative values being decreased. For $R/L \rightarrow \sqrt{3}/2$, however, both curves meet at the same point—the one determined

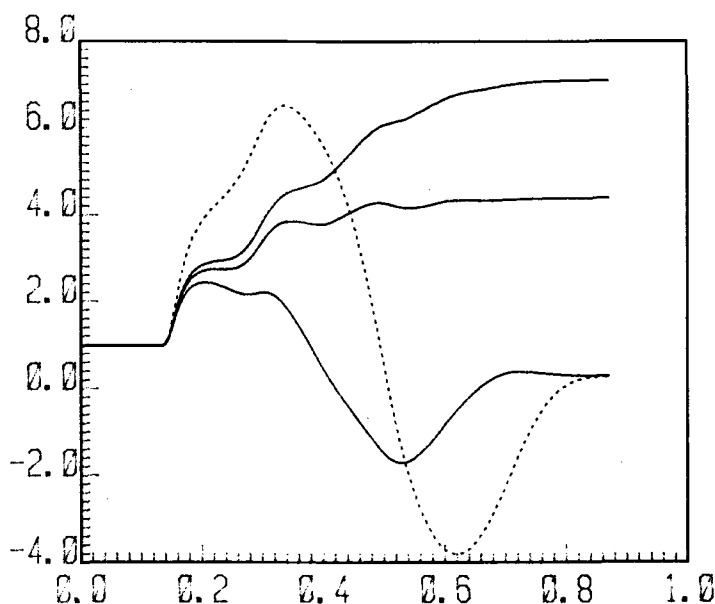


Figure 2. $G_K(R)$, the R -dependent Kirkwood g -factor (average total dipole moment of a sphere of radius R around a reference particle), for the system of figure 1. The symbols have the same meaning as in figure 1.

by (26.2.3D). The curve for $\epsilon_{RF} = \infty$ increases monotonously; with a weaker reaction field ($\epsilon_{RF} = 50$) it remains virtually constant after three shells of neighbours whose dipole moments are positively correlated with the dipole of the reference particle. This is the behaviour one would expect for an infinite system. This is not entirely conclusive, though: first, the dielectric constant used for the reaction field ($\epsilon_{RF} = 50$) is lower than the 'true' dielectric constant ($\epsilon \approx 65$). Second, for $R/L > 1/2$ (the corners of the cube) only incomplete shells of neighbours contribute to $G_K(R)$. We also conclude that, in contrast to what has been said in favour of MI-geometry [40], even local correlations differ more strongly from the infinite system behaviour than those resulting in SC-geometry. Thus SC is not only less time-consuming than MI, but also better.

6. DISCUSSION

It has been rigorously shown that—as has been tacitly assumed for a decade or so—the equations that connect the dipole moment fluctuations of a *spherical* system to its dielectric constant, (1) and (2), are also applicable for the *toroidal* boundary conditions (minimum image, spherical cutoff, reaction field geometry) often used in computer simulations of polar systems. The formula for reaction field geometry with arbitrary ϵ_{RF} is formally identical with that known to hold for periodic boundary conditions (Ewald sum plus reaction field).

The derivation presented above for two and three dimensions is a combination of statistical mechanics and electrostatics. It is completely analogous to the usual Kirkwood–Fröhlich treatment and completely general as well as straightforward. The crucial point is the solution, in Fourier space, of the integral equation for the polarization in the presence of an external field. Due to the close correspondence of that integral equation to the situation typical of computer simulations, the cubical shape of the sample and the toroidal boundary conditions as well as the interaction cutoff and the reaction field may now be treated exactly for the first time.

The only assumptions that had to be made are : (implicitly) that the system be sufficiently large to behave like a macroscopic dielectric (which is unavoidable if macroscopic properties are to be determined); and (explicitly) that the constitutive relation $\mathbf{P} = \chi \mathbf{E}$ be obeyed *locally*, i.e. at every point in space, rather than as a convolution

$$\mathbf{P}(\mathbf{r}) = \int d\mathbf{r}' \chi(\mathbf{r}, \mathbf{r}') \mathbf{E}(\mathbf{r}'), \quad (37)$$

which form might be more appropriate for a small system.

When employed to interpret the results of actual simulations, the fluctuation formulas are found to give a consistent estimate of the dielectric constant for RF-geometry, while, for medium-size systems, the distortions of local correlations induced by MI or SC-geometry render those boundary conditions impractical. (This may apply to measurements of the polarization induced by an external field too.) In the thermodynamic limit the dielectric constant of the system studied here seems to be well above the values reported for EK-geometry in the literature.

The N -dependence of the dielectric constant, and the apparent inconsistency between its RF and EK estimates in particular, will be investigated in more detail. Because of the enormous amount of computer time involved comments from, and co-operation with, other laboratories would be most welcome.

The integral equation defining the macroscopic polarization may be used to study the influence of various geometries on the dipolar field of a particle. An investigation of this kind has been completed. Results will be reported in a separate publication [37].

The simulations reported in § 5 were carried out on a CDC CYBER 170-720 at the Interuniversitäres EDV-Zentrum and on a PDP 11/34 at the Prozeß-rechenanlage Physik, both at Vienna. Many of the ideas presented in this article began to take shape while I was at the Vitreous State Laboratory. For making possible my stay there I would like to thank Professor T. A. Litovitz as well as Dr. H. A. Posch and Professor P. Weinzierl. It is a special pleasure to acknowledge the stimulating discussions I had with Dr. O. Steinhauser.

APPENDIX

Convolution with toroidal boundary conditions

Because of the importance of (15) in the text we will show here explicitly, for the one-dimensional case, that

$$\tilde{h}(k) = \tilde{f}(k)\tilde{g}(k) \quad (\text{A } 1)$$

is the Fourier transform of the convolution integral

$$h(x) = \int_{\text{TBC}} dx' f(x')g(x-x') \quad (\text{A } 2)$$

with toroidal boundary conditions. f , g , and h are assumed to be defined on the interval $(-1/2, 1/2)$; k is a point in Fourier space that is compatible with the length of this interval, i.e. k is a multiple of 2π .

Let $0 < x < 1/2$. f and g are only defined on $(-1/2, 1/2)$, so that (A 2), written explicitly, now reads

$$h(x) = \int_{-\frac{1}{2}}^{x-\frac{1}{2}} dx' f(x')g(x-x'-1) + \int_{x-\frac{1}{2}}^{\frac{1}{2}} dx' f(x')g(x-x'). \quad (\text{A } 3)$$

Inserting the Fourier expansion for f and g ,

$$\begin{aligned} h(x) &= \int_{-\frac{1}{2}}^{x-\frac{1}{2}} dx' \left[\sum_{k'} \exp(ik'x')\tilde{f}(k') \right] \left[\sum_k \exp[ik(x-x'-1)]\tilde{g}(k) \right] \\ &\quad + \int_{x-\frac{1}{2}}^{\frac{1}{2}} dx' \left[\sum_{k'} \exp[ik'x']\tilde{f}(k') \right] \left[\sum_k \exp[ik(x-x')]\tilde{g}(k) \right] \\ &= \int_{-\frac{1}{2}}^{x-\frac{1}{2}} dx' \sum_{k,k'} \exp i(k'-k)x' \exp(ikx) \exp(-ik)\tilde{f}(k')\tilde{g}(k) \\ &\quad + \int_{x-\frac{1}{2}}^{\frac{1}{2}} dx' \sum_{k,k'} \exp[i(k'-k)x'] \exp(ikx)\tilde{f}(k')\tilde{g}(k), \end{aligned} \quad (\text{A } 4)$$

and since $\exp(-ik) = 1$ for every k , both integrals may be combined :

$$h(x) = \sum_{k,k'} \exp(ikx)\tilde{f}(k')\tilde{g}(k) \int_{-\frac{1}{2}}^{\frac{1}{2}} dx' \exp[i(k'-k)x']. \quad (\text{A } 5)$$

The remaining integral is just $\delta_{k,k'}$, so that

$$h(x) = \sum_k \exp(ikx)\tilde{f}(k)\tilde{g}(k), \quad (\text{A } 6)$$

which proves (A 1).

For $-1/2 < x < 0$ the proof is similar.

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